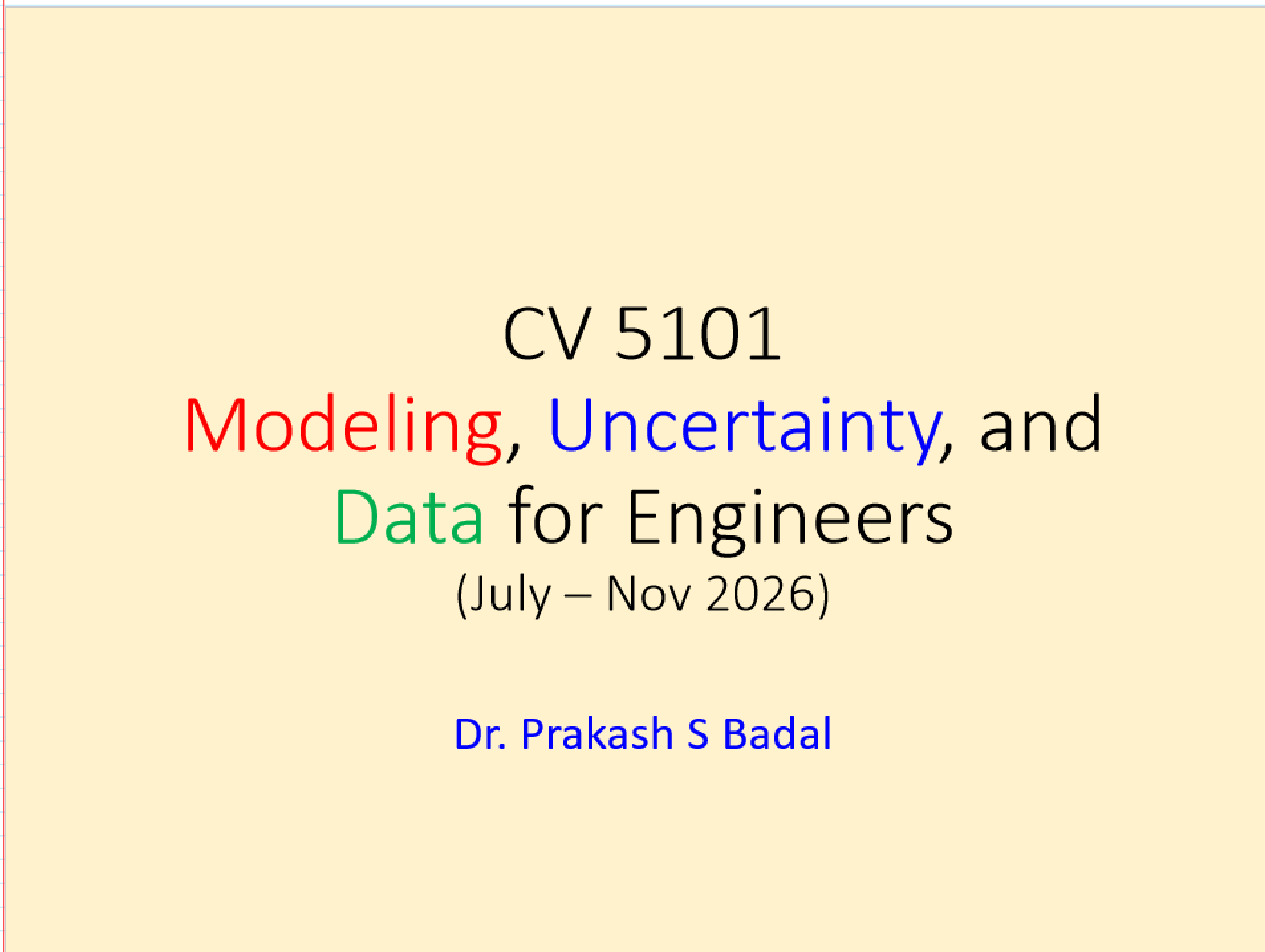
inClass-P2-L1b



Flow

- Module Outline
- Motivation
- Introduction

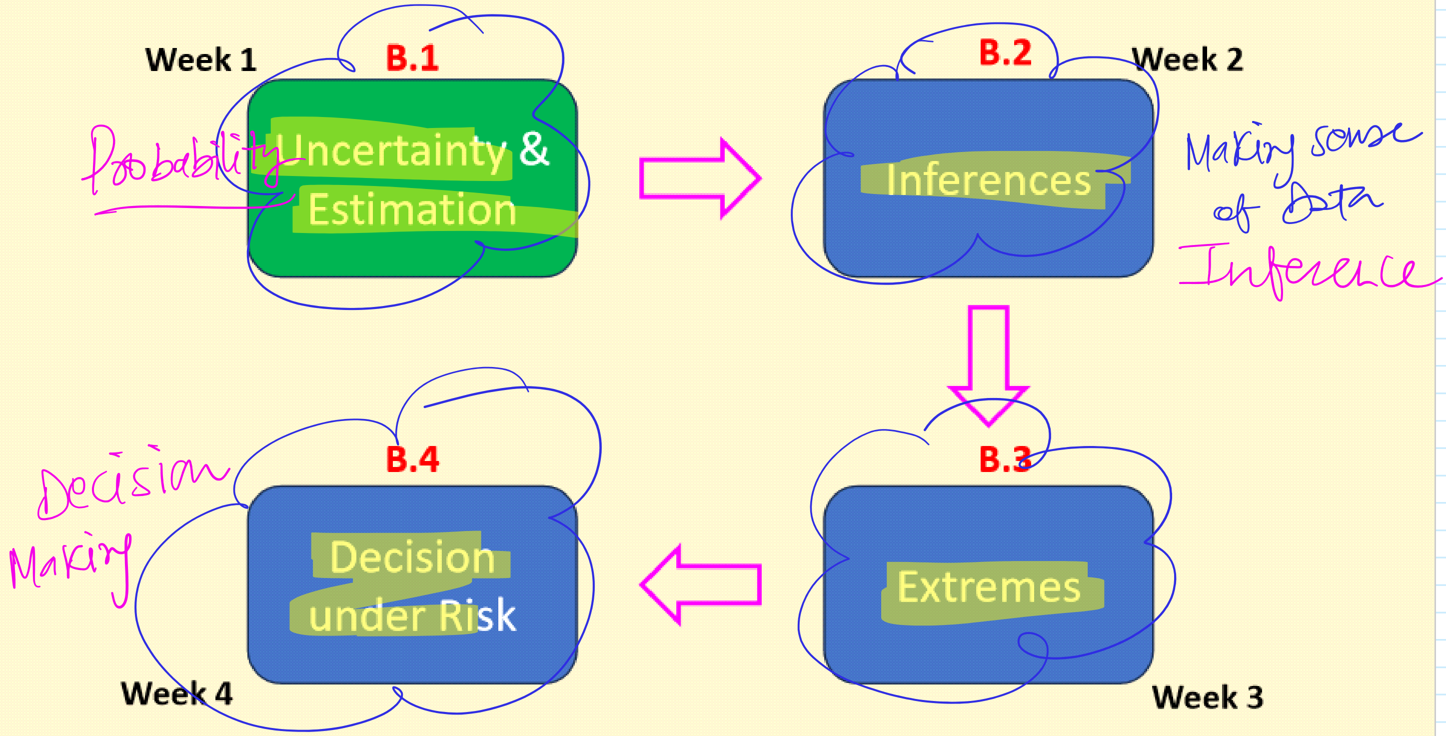
Announcement

- Usual class hours: T (5 pm), W (11 am), R (9 am), F (8 am)
- Labs on Mondays at 2 pm
- Generally, no classes on Fridays
- Part-2 until Oct 5
- Quiz-2 on Oct 13
- Check github regularly

September 2026

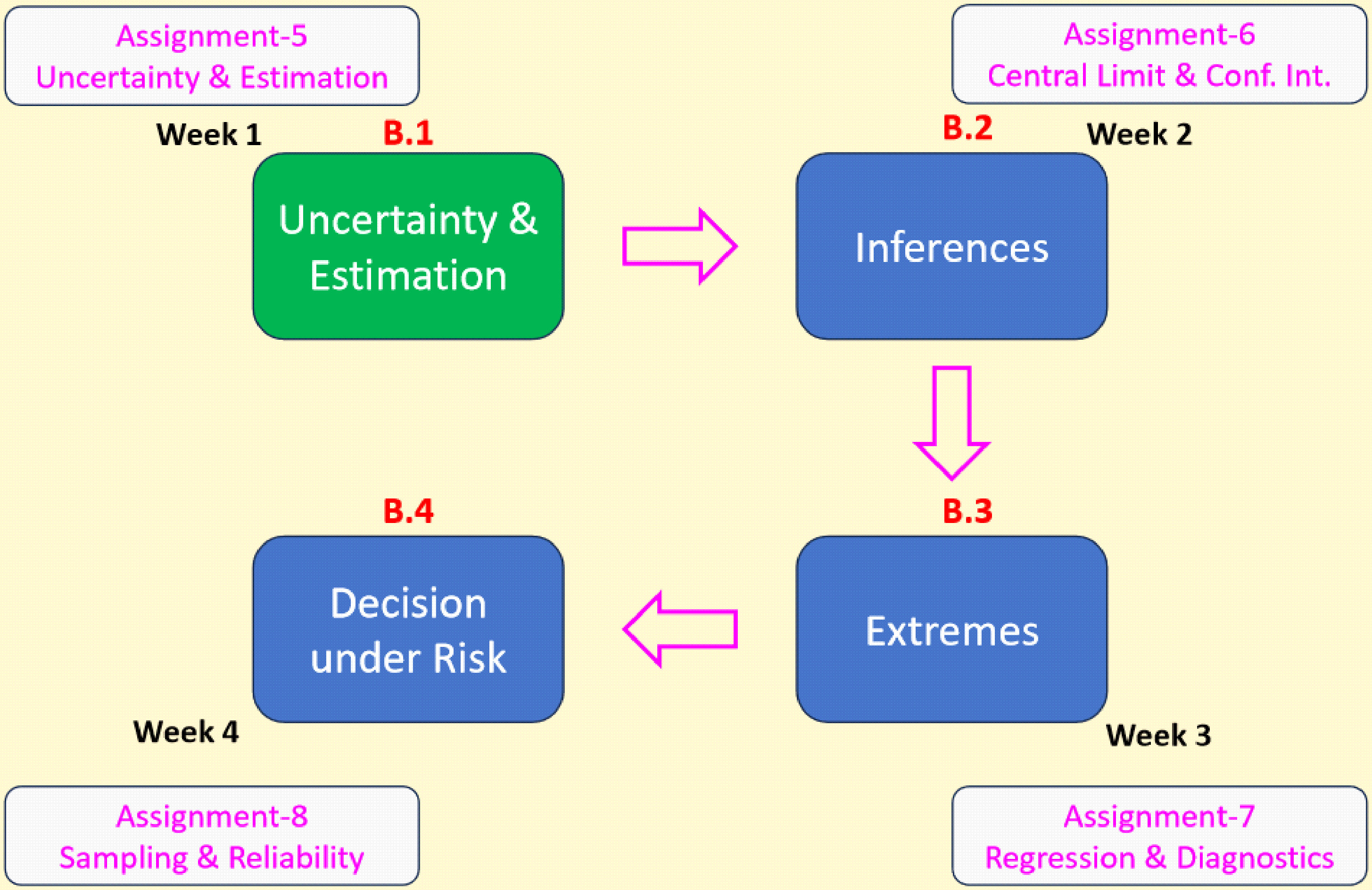
SUN	MON	TUE	WED	THU	FRI	SAT
		1	2	3	4 F&G swapped. Class at 9 am.	5
6	7 Assmt-5	8	9	10	11	12
13	14 Holiday	15 OOT	16	17	18	19
20	21 Assmt-6	22	23	24	25	26
27	28 Assmt-7	29	30	1 Oct	2 Oct Holiday	
	5 Oct Assmt-8					

Module Overview



A crash course in probability, probabilistic models, and probabilistic methods.

Module Overview



Module Overview

B.1

Uncertainty & Estimation

B.1.1

Probabilistic basics and RVs
- Probability laws
- Random variables
- Discrete, continuous RVs
- PMF/PDF/CDF
- Expectation
- Variance/cov./corr.

Q1C5, Funda6-7-8

B.1.2

Core distributions
- Gaussian
- Uniform
- Exponential
- Lognormal
- Symmetric/skewness
- *Tails*: thick/light

Q1C5, Funda9-10

B.1.3

Propagation laws & least-sq.
- Uncertainty propagation
- Least-square errors
- Parameter estimation

Q1C2-C3, Funda9-10

Module Overview

B.2

Inferences

B.2.1

Confidence interval (CI)

- Interval estimation
- Mean, stdev

Q1C3.5

B.2.2

Hypothesis testing (HT)

- p-value
- Null/alternative hypothesis
- Type I/II error
- Power

Q1C3.9

B.2.3

Goodness-of-Fit (GoF).

- Likelihood ratio
- KS test

Q1C5.4

Module Overview

Extremes & Risk Analysis

B.3

Extreme value
- Gumbel dist. (Q1C5.3)
- GEV dist. (generalized)
- Return period
- Design life
- Prob. of failure

Q2C7

B.4

Probabilistic risk analysis
- Hazard function
- Fragility function
- Risk assessment

(Not from teach book)

Check-in with teach-book

https://mude.citg.tudelft.nl/book/2024/

- Q1 Topics (Chapters 5, 6, 2, and 3)
  - Q1C5 Univariate continuous distribution
    - Q1C5.1 PDF/CDF
    - Q1C5.2 Empirical Distributions
    - Q1C5.3 Parametric Distributions
    - Q1C5.4 Fitting a Distribution
  - Q1C6 Multivariate Distributions (briefly)
  - Q1C2 Propagation of Uncertainty
  - Q1C3 Observation Theory: least-sq., Hyp. Test, Conf. Intervals
- Q2 Topics (Chapter 7 and 8)
  - Q2C7 Extreme value theory: GEV, return period, POT
  - Q2C8 Risk and decision making (CBA)
- Fundamental Concepts
  - Chapter 6, 7, 8, 9. Probability basics, rv, z- and t-tables
- Programming
  - Fundamental Concepts → Chapter 10

Questions, comments, or concerns?

10